Assesment - 4

classmate
Date
Page

- **Aim :-**
  - To write and simulate verilog codes SR, JK, D and T flip flops.
  - To understand their working and verify outputs through simulation.

- **Theory :-**

1. Combinational (VS) Sequential Circuits
2. Latch (VS) flip flops
3. Triggering signal in Sequential circuits
4. flip-flops.

- **Tools Required :-**

- Software : Xilinx ISE / Vivado / ModelSim
- Language : Verilog HDL
- Hardware : FPGA Kit
- Computer System with simulation environment.

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23-Sep-25

Assesment - 4

- SR - flip flop :-

```
module SR_ff (clk, rst, s, r, q);
input clk, rst, s, r;
output reg q;
always @ (posedge clk or posedge rst)
begin
if (rst == 1)
 q <= 0;
else if (s == 0 && r == 0)
q <= q;
else if (s == 0 && r == 1)
 q <= 0;
else if (s == 1 && r == 0)
q <= 1;
else if (s == 1 && r == 1)
q <= 1'bx;
end
endmodule
```

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```
module tb ();
reg clk, rst, s, r;
wire q;
SR_ff out (clk, rst, s, r, q);
always begin
#10 clk = ~clk;
end
initial
begin
clk = 0; rst = 0; s = 0; r = 0;
#5 rst = 1;
#5 rst = 0; s = 1; r = 0;
#20 s = 0; r = 1; s = 0; r = 0;
#20 rst = 1; rst = 0;
end
```

JK - flip flop

```
module JK_ff (clk, rst, j, k, q);
input clk, rst, j, k;
output reg q;
always @(posedge clk or posedge rst)
begin
if (rst == 1)
q <= 0;
else if (j == 0 && k == 0)
q <= q;
else if (j == 0 && k == 1)
q <= 0;
else if (j == 1 && k == 0)
q <= 1;
else if (j == 1 && k == 1)
q <= ~q;
end
endmodule
```

```
module tb();
reg clk, rst, j, k;
wire q;
jk_ff uut (clk, rst, j, k, q);
always begin
#10 clk = ~clk;
end
initial
begin
clk = 0; rst = 0; j = 0; k = 0;
#5 rst = 1;
#5 rst rst = 0; j = 0; k = 0;
#25 j = 0; k = 1;
#25 j = 1; k = 1;
#25 j = 1; k = 0;
end
endmodule
```

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T-flip flop

```
module t_ff (clk, rst, t, q);
input clk, rst, t;
output reg q;
always @ (posedge clk or posedge rst)
 begin
if (rst == 1)
 q <= 0;
else if (t == 0)
q <= q;
else
q <= ~q;
end
endmodule
```

```
module tb ();
reg clk, rst, t;
wire q;
t_ff uut (clk, rst, t, q);
always begin
#10 clk = ~clk;
end
initial
begin
clk = 0; rst = 0; t = 0;
#5 rst = 1;
#5 rst = 0; t = 0;
#15 t = 1;
#10 t = 0;
#15 t = 1;
#5 rst = 1;
#5 rst = 0; t = 1;
 $stop
 end
endmodule
```

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D-flip flop

```
module d_ff(clk, rst, d, q);
input clk, rst, d;
output reg q;
always @(posedge clk or posedge rst)
begin
if(rst==1)
q<=0;
else
q<=d;
end
endmodule
```

```
module tb();
reg clk, rst, d;
wire q;
d_ff uut(clk, rst, d, q);
always begin
#10 clk=~clk;
end
initial
 begin
clk=0; rst=0; d=0;
#5 rst=1;
#5 rst=0; d=0;
#15 d=1;
#ld d=0;
#15 d=1;
#5 rst=1;
#5 rst=0; d=1;
$stop
end
endmodule
```

o/p verified

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24BCE0403

(*) Answer the following :-

(1) Difference blw combinational And Seawential Circuits.

Ans Combinational :- output depends only on present inputs (no memory)

Seawential :- output depends on present input and past history. (uses memory).

(2) Difference blw latch and flip flop.

Ans latch :- level - triggered (works when input / clock is at a level).

flip flop :- Edge - triggered (works only on clock transition - rising or falling edge).

(3) what are the different triggering signal in seawential circuits.

- positive edge Triggering (rising edge of clock)
- Negative Edge Triggering (falling edge of Clock)
- level Triggering (high level or low level of clock)

ModelSim - INTEL FPGA STARTER EDITION 2020.1

Activities Vsim

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Search:

sim - Default

Instance	Design unit	Design
ttt_tbff	ttt_tbff	Module
uut	ttt_ff	Module
#ALWAYS#19	ttt_tbff	Process
#INITIAL#22	ttt_tbff	Process
#vsim_capacity#		Capacity

Wave - Default

Msgs
/ttt_tbff/clk 0
/ttt_tbff/rst 1
/ttt_tbff/t 0
/ttt_tbff/q St0

Now 60 ps

Cursor 1 7 ps

Project sim tff.v Wave

Transcript

```
# ** Note: $stop    : /home/matlab/24BCB0100_Charishma/tff.v(32)
#    Time: 60 ps  Iteration: 0  Instance: /ttt_tbff
# Break in Module ttt_tbff at /home/matlab/24BCB0100_Charishma/tff.v line 32
VSIM 2>
```

0 ps to 128 ps Project: engbuddh Now: 60 ps Delta: 0 /ttt_tbff/t

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ModelSim - INTEL FPGA STARTER EDITION 2020.1

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ModelSim - INTEL FPGA STARTER EDITION 2020.1

C:/intelFPGA/18.1/asd.v (/tb) - Default

Ln#

```
1   module d_ff(clk,rst,d,q);
2   input clk,rst,d;
3   output reg q;
4   always@(posedge clk or posedge rst)
5   begin
6   if(rst==1)
7   q<=0;
8   else
9   q<=d;
10  end
11  endmodule
12
13  module tb();
14  reg clk,rst,d;
15  wire q;
16  d_ff uut(clk,rst,d,q);
17  always begin
18  #10 clk=~clk;
19  end
20  initial
21  begin
22  clk=0;rst=0;d=0;
23  #5 rst=1;
24  #5 rst=0;d=0;
25  #15 d=1;
26  #10 d=0;
27  #15 d=1;
28  #5 rst=1;
29  #5 rst=0; d=1;
30  $stop;
31  end
32  endmodule
```

C:/intelFPGA/18.1/asd.v (/tb) - Default

```
Ln#
 1  module sr_ff(clk,rst,s,r,q);
 2  input clk,rst,s,r;
 3  output reg q;
 4  always@(posedge clk or posedge rst)
 5  begin
 6  if(rst==1)
 7  q<=0;
 8  else if (s==0 && r==0)
 9  q<=q;
10  else if (s==0 && r==1)
11  q<=0;
12  else if (s==1 && r==0)
13  q<=1;
14  else if (s==1 && r==1)
15  q<=1'bx;
16  end
17  endmodule
18
19  module tb();
20  reg clk,rst,s,r;
21  wire q;
22  sr_ff uut(clk,rst,s,r,q);
23  always begin
24  #10 clk=~clk;
25  end
26  initial
27  begin
28  clk=0;rst=0;s=0;r=0;
29  #5 rst=1;
30  #5 rst=0;s=1; r=0;
31  #20 s=0; r=1;
32  #20 s=1; r=1;
33  #20 s=0; r=0;
34  #20 rst=1;
35  #5 rst=0;
36  $stop;
37  end
38  endmodule
39
```

C:/intelFPGA/18.1/asd.v (/tb) - Default

```
Ln#
 1  module t_ff(clk,rst,t,q);
 2  input clk,rst,t;
 3  output reg q;
 4  always@(posedge clk or posedge rst)
 5  begin
 6  if(rst==1)
 7  q<=0;
 8  else if (t==0)
 9  q<=q;
10  else
11  q<=~q;
12  end
13  endmodule
14  module tb();
15  reg clk,rst,t;
16  wire q;
17  t_ff uut(clk,rst,t,q);
18  always begin
19  #10 clk=~clk;
20  end
21  initial
22  begin
23  clk=0;rst=0;t=0;
24  #5 rst=1;
25  #5 rst=0;t=0;
26  #15 t=1;
27  #10 t=0;
28  #15 t=1;
29  #5 rst=1;
30  #5 rst=0; t=1;
31  $stop;
32  end
33  endmodule
34
```

C:/intelFPGA/18.1/asd.v (/tb) - Default

```
Ln#
 1  module jk_ff(clk,rst,j,k,q);
 2  input clk,rst,j,k;
 3  output reg q;
 4  always@(posedge clk or posedge rst)
 5  begin
 6  if(rst==1)
 7  q<=0;
 8  else if (j==0 && k==0)
 9  q<=q;
10  else if (j==0 && k==1)
11  q<=0;
12  else if (j==1 && k==0)
13  q<=1;
14  else if (j==1 && k==1)
15  q<=~q;
16  end
17  endmodule
18  module tb();
19  reg clk,rst,j,k;
20  wire q;
21  jk_ff uut(clk,rst,j,k,q);
22  always begin
23  #10 clk=~clk;
24  end
25  initial
26  begin
27  clk=0;rst=0;j=0;k=0;
28  #5 rst=1;
29  #5 rst=0;j=0; k=0;
30  #25 j=0; k=1;
31  #25 j=1; k=1;
32  #25 j=1; k=0;
33  #5 rst=1;
34  #25 rst=0; j=1; k=1;
35  $stop;
36  end
37  endmodule
38
```